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Digital Image Processing

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CONTENTS



1

IMAGE RESTORATION (2)

2

IMAGE COMPRESSION

3

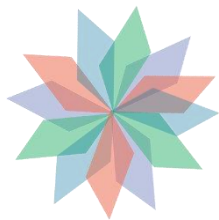
IMAGE RECONSTRUCTION



IMAGE RESTORATION (2)

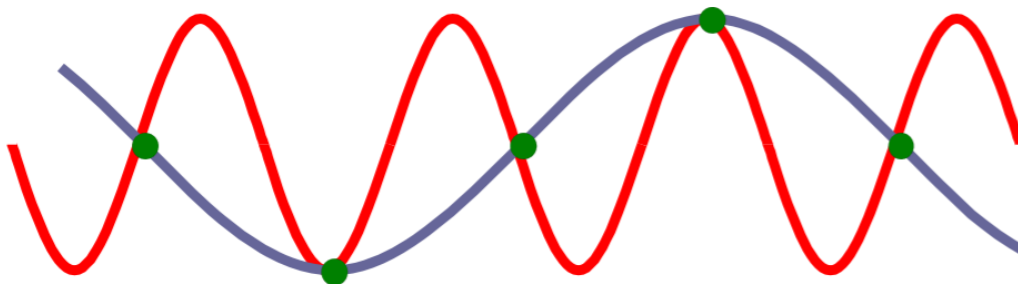


Aliasing



Aliasing

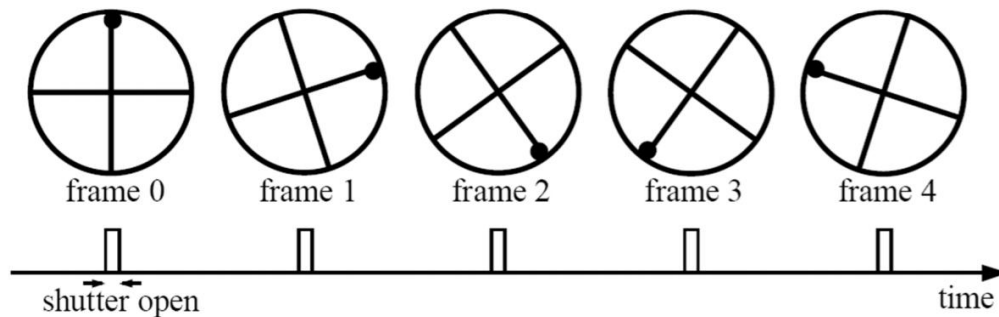
- ▶ Aliasing means that signal “traveling in disguise” as other frequency



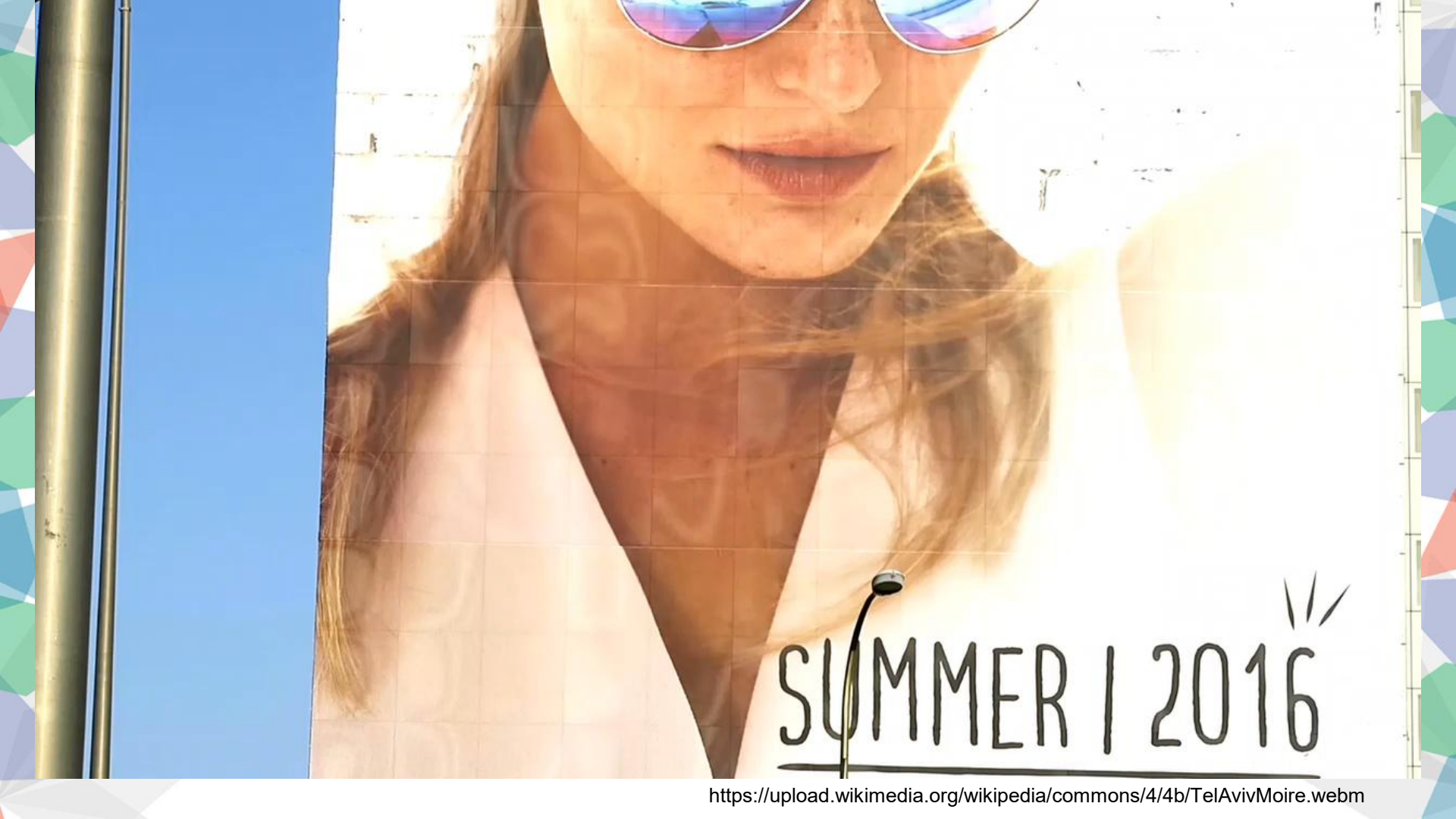
- ▶ The 2 frequencies above cannot be distinguished due to insufficient samples

Aliasing

- ▶ In animation or video, there are many examples for aliasing



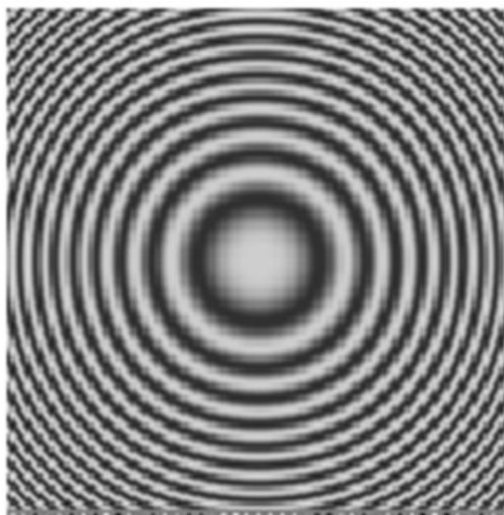
- ▶ Without dot, wheel appears to be rotating backward instead of forward



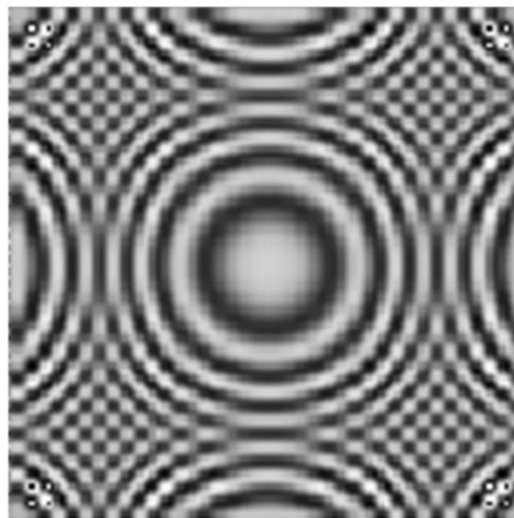
SUMMER | 2016

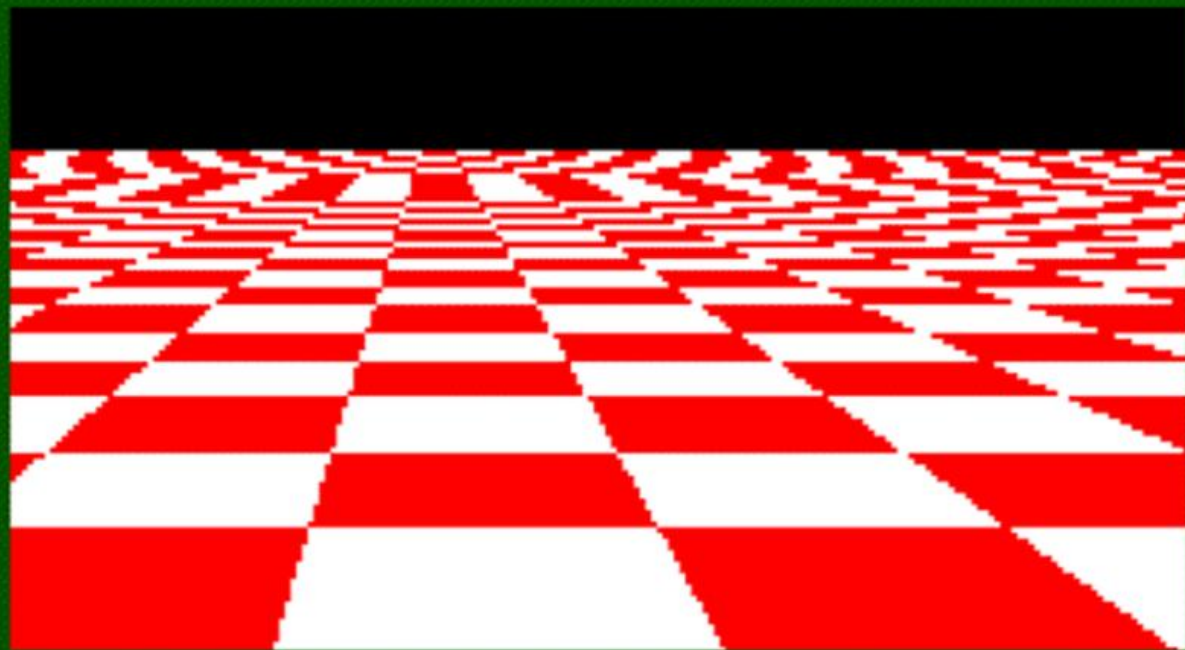
Aliasing in 2D images

original



reconstruction
(MOIRE PATTERN)



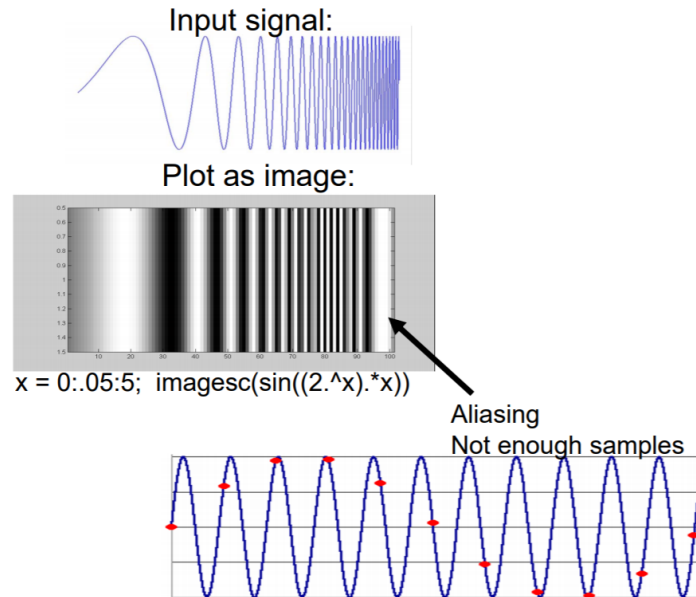


Disintegrating textures

Aliasing in 2D images



Aliasing in 2D images



Anti-Aliasing

- ▶ Increase sampling frequency
 - ▶ E.g. Render 4 rays per pixel
- ▶ Reduce maximum frequency
 - ▶ E.g. Lowpass filter before sampling

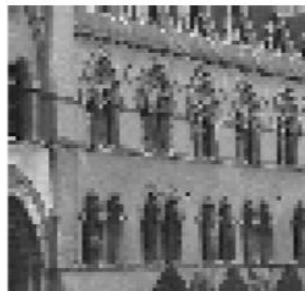
Example



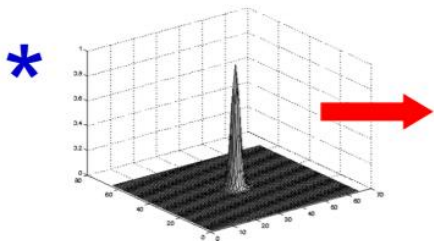
down sample by
factor of 4



4 x zoom



convolve with
Gaussian



down sample by
factor of 4





Image Scaling Algorithms

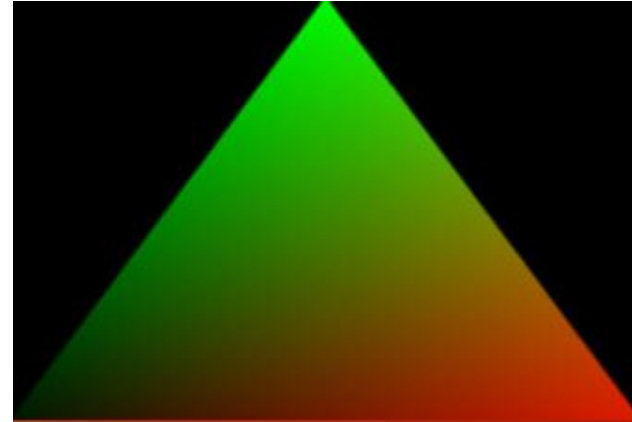
Image Scaling

- ▶ Image scaling is one of common processes in digital image processing
- ▶ However, using a raster (pixel) graphics image, the new image most likely loses its quality
 - ▶ When the image is scaled up, the number of pixels increases.
 - ▶ When the image is scaled down, the number of pixels decreases.
- ▶ Both cause aliasing

```
scale_x = new_width / width  
scale_y = new_height / height
```

Interpolation

- ▶ Interpolation is used for quickly and accurately estimating values without defining them
- ▶ Interpolation uses the estimation of value between two or more distinct points
- ▶ Interpolation can also be used for scale the image

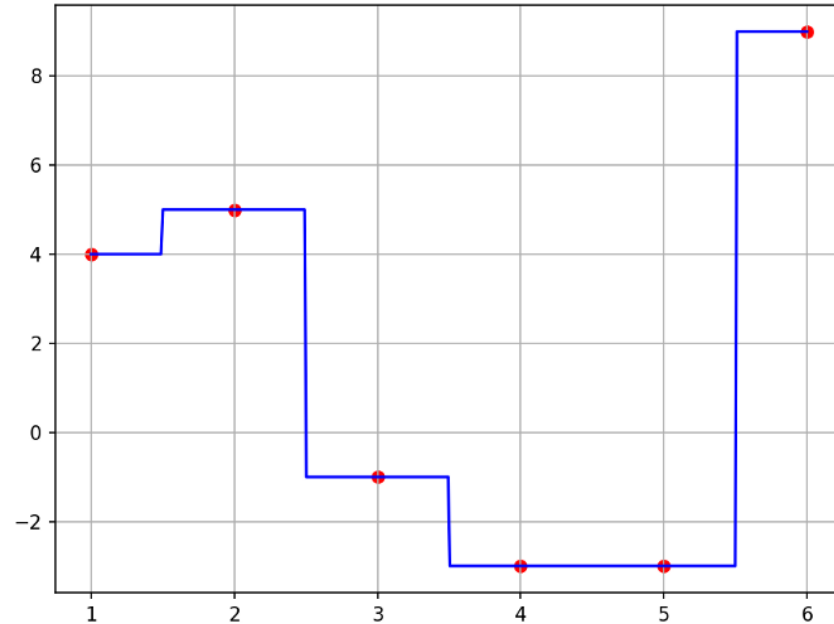


Nearest-neighbor interpolation

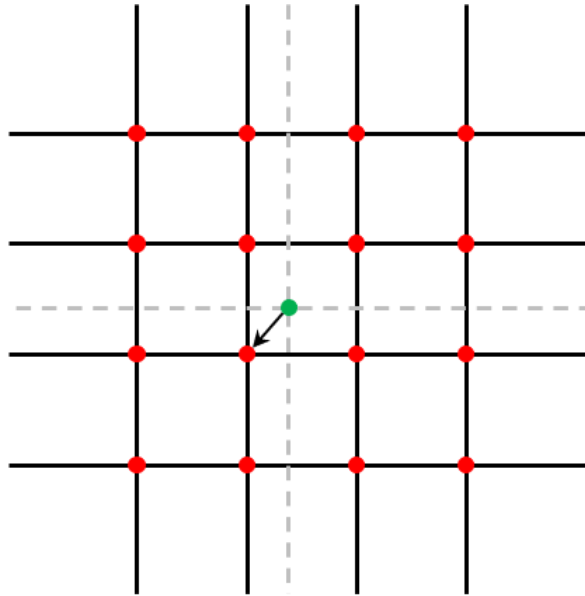
```
scale_x = new_width / width  
scale_y = new_height / height
```

- ▶ One of the simplest methods of image scaling
- ▶ After image is scaled, the value of the output pixel will be the nearest pixel in the input
- ▶ The detail of the edge will be sharp for the pixel art, but the image will be jagged in smoothed input

Nearest-neighbor interpolation



Nearest-neighbor interpolation



Nearest-neighbor interpolation



**Pixel art processed using
Nearest-neighbor interpolation**



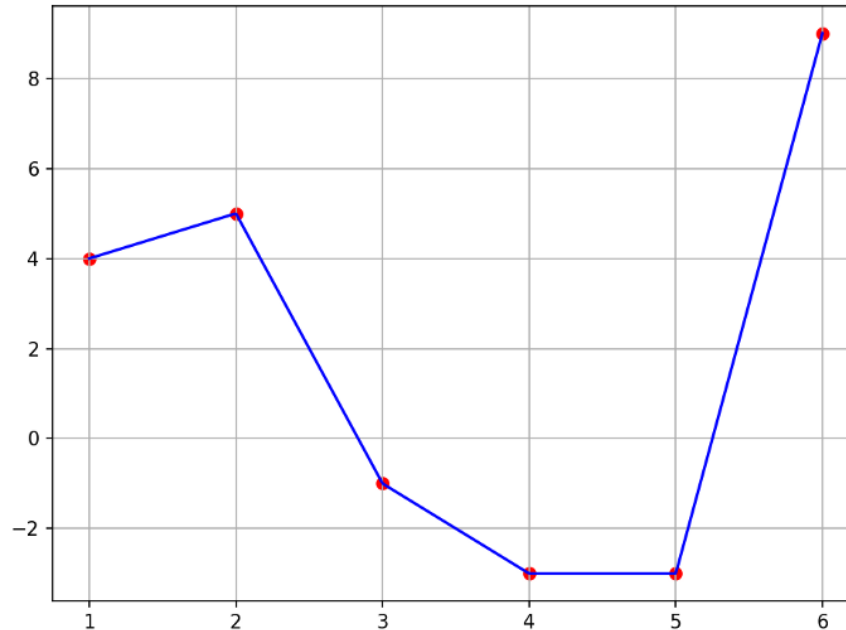
**Smoothed image processed using
Nearest-neighbor interpolation**

Bilinear interpolation

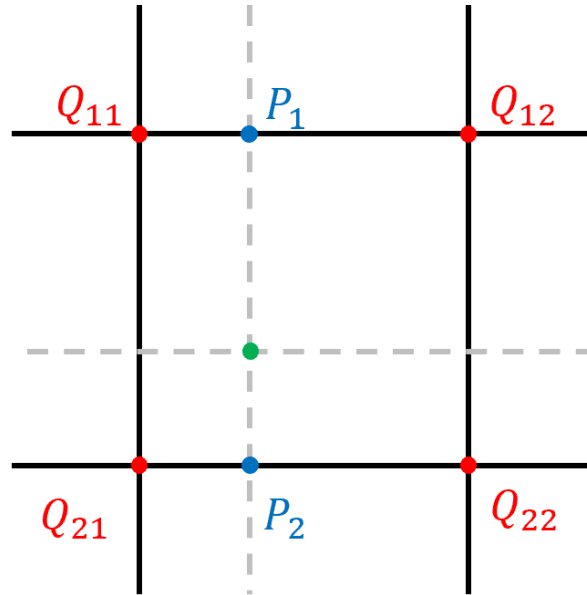
```
scale_x = new_width / width  
scale_y = new_height / height
```

- ▶ This method interpolates the colour values using linear transition
- ▶ After image is scaled, the value of the output pixel will be the weighted average of the neighbour pixels by their distance
- ▶ The detail will be fine for smoothed input but reduces the sharp of the edges which also reduces the output contrast

Bilinear interpolation



Bilinear interpolation



Bilinear interpolation



**Pixel art processed using
Bilinear interpolation**



**Smoothed image processed using
Bilinear interpolation**

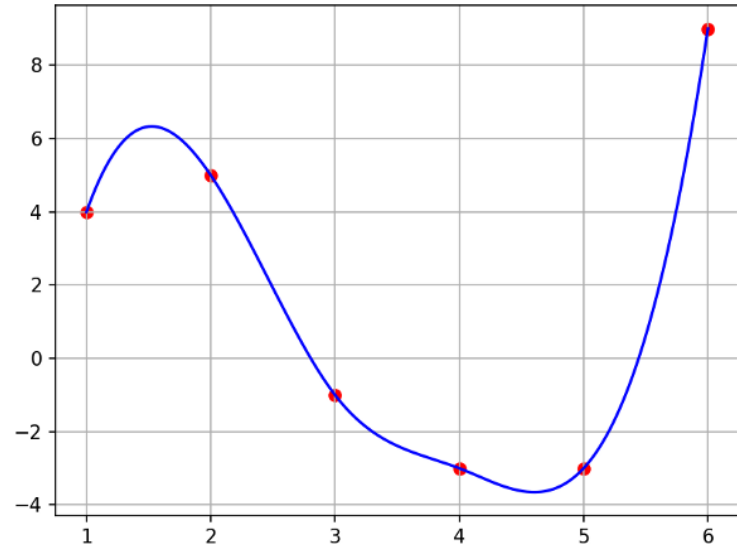
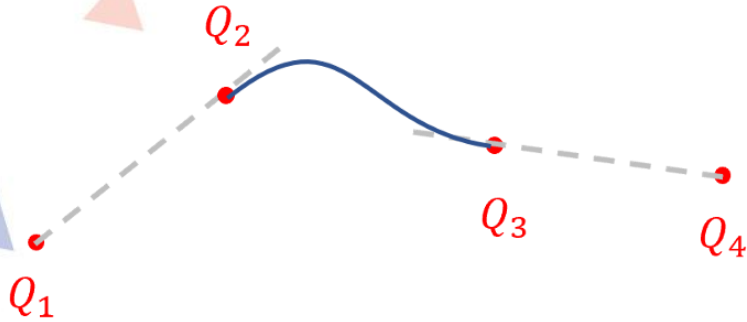
Bicubic interpolation

```
scale_x = new_width / width  
scale_y = new_height / height
```

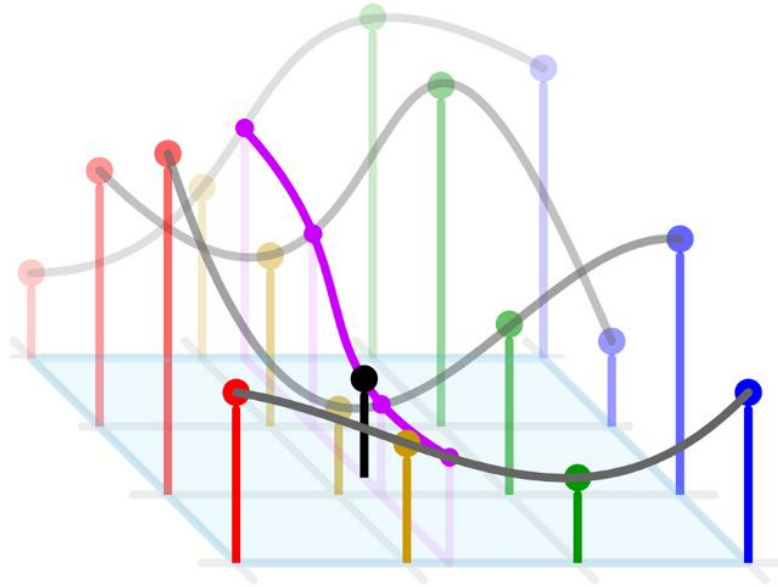
- ▶ This method interpolates the colour values using bicubic spline transition
- ▶ The detail is similar to the bilinear interpolation, but the computation cost is subsequently higher

$$f(x) = ax^3 + bx^2 + cx + d$$

Bicubic interpolation



Bicubic interpolation



Bicubic interpolation

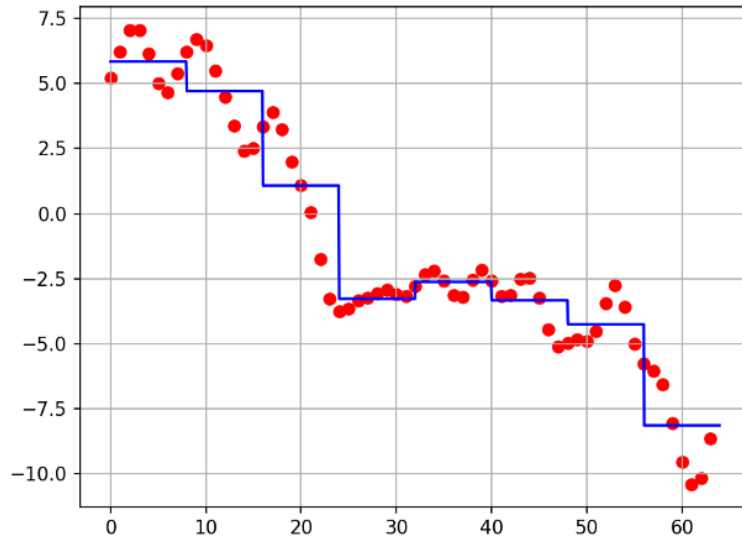


**Pixel art processed using
Bicubic interpolation**



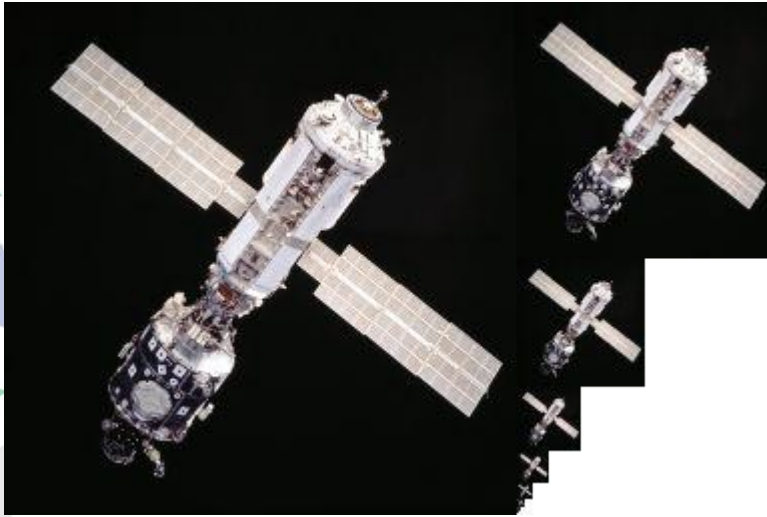
**Smoothed image processed using
Bicubic interpolation**

Other image scaling algorithms – Box sampling



- ▶ Specifically for downscaling an image
- ▶ The output pixel value is the average of the colour inside the downsampled box
- ▶ The weakness is the computational cost

Other image scaling algorithms – MIPmap



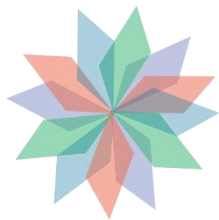
- ▶ Using pre-calculated downscaled images to reduce the calculation time and loss of quality might occur, but increase the storage space for more images
- ▶ Used often in Level of Detail algorithm and other fast rendering algorithms
- ▶ MIP is from Latin phrase **multum in parvo** meaning “**much in little**”

Other image scaling algorithms

- ▶ Sinc resampling
 - ▶ Best result in theory, cannot be fully computed in digital images
- ▶ Lanczos resampling
 - ▶ Approximation to the sinc resampling
- ▶ Vectorization
 - ▶ Trace the edges into vectors
- ▶ Deep convolutional neural networks
 - ▶ Using machine learning to scale the images



Super Resolution



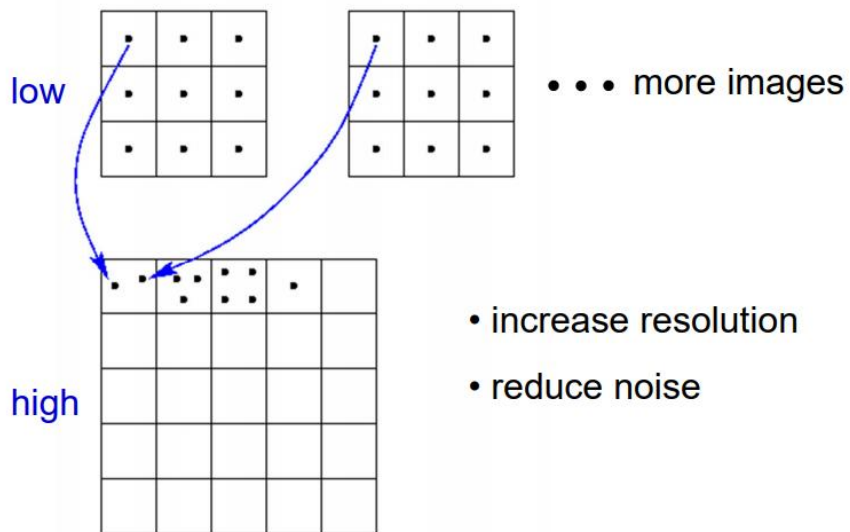
Super Resolution

Suppose there are multiple images of the same scene
each displaced spatially ...

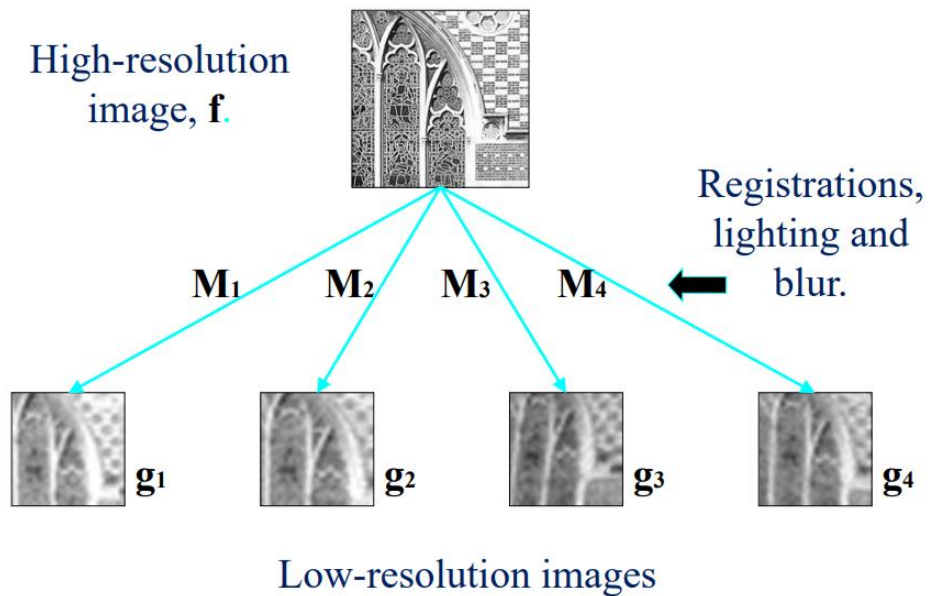


Super Resolution

Treat images as point samples



Super Resolution



Super Resolution

Write the generative model for one image i as

$$\mathbf{g}_i = \mathbf{M}_i \mathbf{f} + \boldsymbol{\eta}_i$$

where $\mathbf{M}_i$ combines registration, lighting and down-sampling.
Finally the generative models of all N images are stacked vertically to form an over-determined linear system

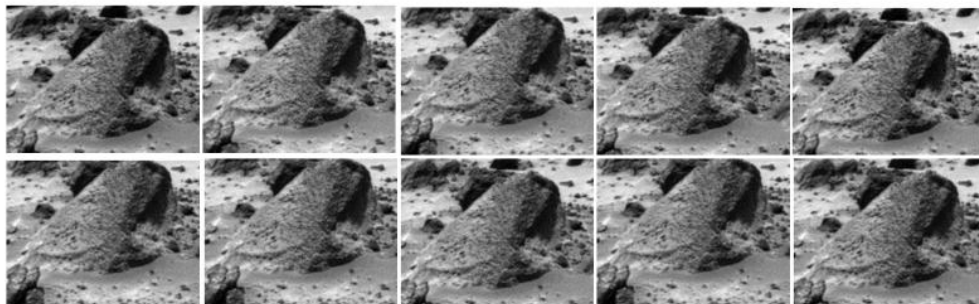
$$\begin{bmatrix} \mathbf{g}_0 \\ \mathbf{g}_1 \\ \vdots \\ \mathbf{g}_{N-1} \end{bmatrix} = \begin{bmatrix} \mathbf{M}_0 \\ \mathbf{M}_1 \\ \vdots \\ \mathbf{M}_{N-1} \end{bmatrix} \mathbf{f} + \begin{bmatrix} \boldsymbol{\eta}_0 \\ \boldsymbol{\eta}_1 \\ \vdots \\ \boldsymbol{\eta}_{N-1} \end{bmatrix}$$

The MAP estimator has the form:

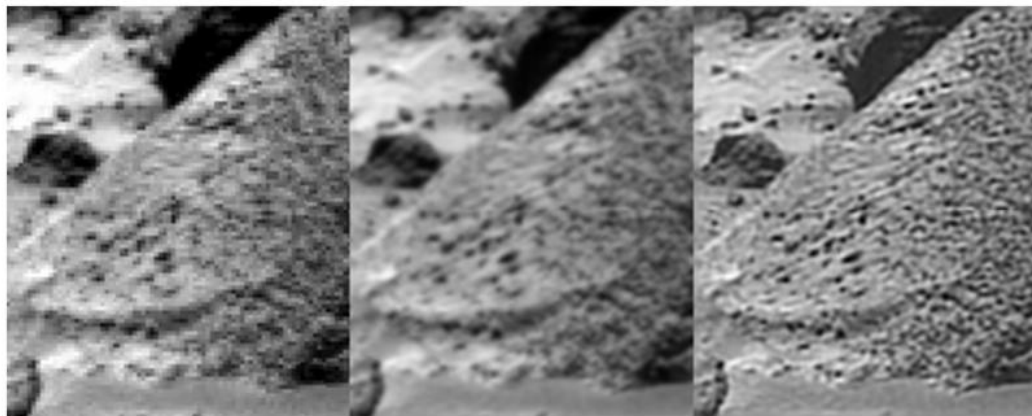
$$\mathbf{f}_{\text{MAP}} = \underset{\mathbf{f}}{\operatorname{argmin}} \left(\underbrace{(\mathbf{g} - \mathbf{M}\mathbf{f})^2}_{\text{likelihood}} + \lambda^2 \underbrace{p(\mathbf{f})}_{\text{prior}} \right)$$

Super Resolution

25 JPEG images courtesy of the Mars lander
images are from different sweeps of a rotating camera



Super Resolution



Original frame

Average image

Super-resolution

2x zoom from 25 JPEG images.

Super Resolution



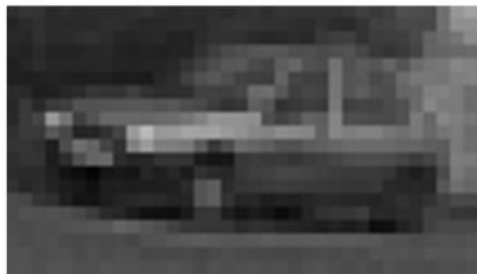
Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Super Resolution



Super Resolution

85 JPEG images



original ROI

35 x 20 pixels



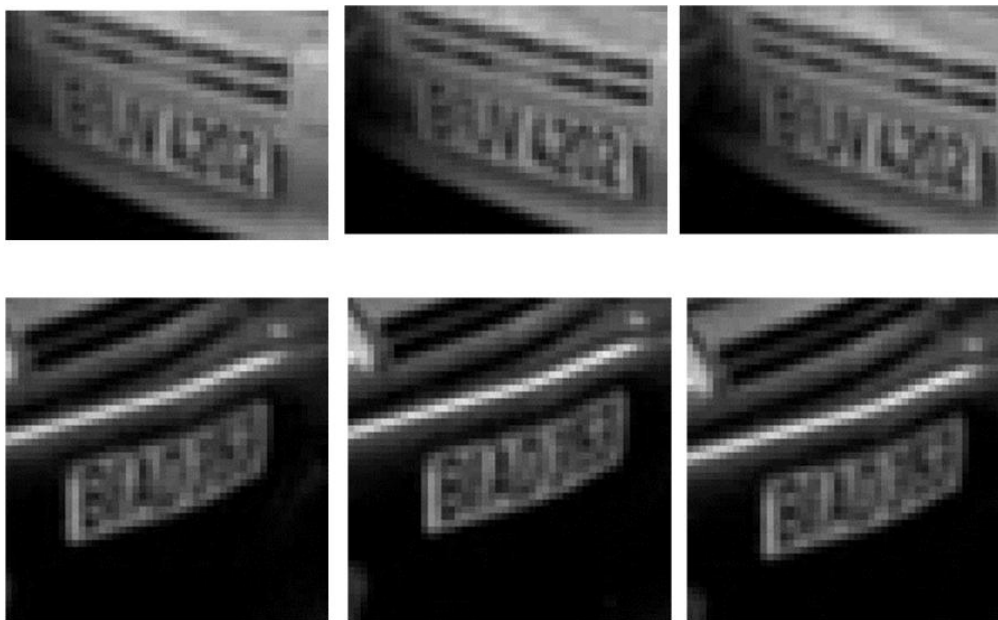
four times resolution

Super Resolution



Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Super Resolution



Super Resolution





IMAGE COMPRESSION

Image Compression

- ▶ Image compression is one of data compression algorithms to reduce the size which can save the cost of storage and transmission
- ▶ The image compression can be separated into 2 main types
 - ▶ Size compression
 - ▶ Using image scaling algorithms
 - ▶ Can be easily notice the differences
 - ▶ Colour compression
 - ▶ Using image lossy compression
 - ▶ Hard to notice the differences due to human visual perception

Lossless and lossy image compression

- ▶ Lossless compression is the algorithms which allows the original data to be able to be reconstructed from the compressed data without losing any information, for example, ZIP files
 - ▶ In image – GIF, PNG, TIFF
- ▶ Lossy compression is the algorithms which some information is loss in exchange of more size reduction
 - ▶ In image JPEG, HEIF
- ▶ Some file formats can be either lossless or lossy
 - ▶ JPEG 2000, JPEG XR, WebP

Lossless image compression

- ▶ There are many types of lossless image compression
 - ▶ Chain Codes
 - ▶ Huffman Coding
 - ▶ Arithmetic Coding
 - ▶ Run-length Encoding

Lossy image compression

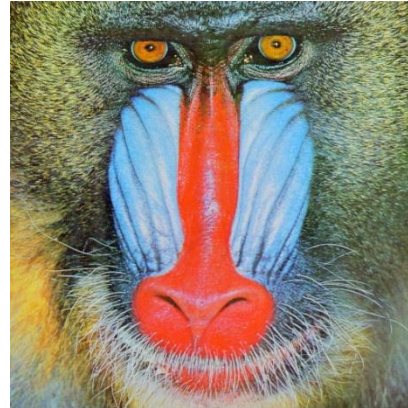
- ▶ There are many types of lossy image compression
 - ▶ Transform Coding
 - ▶ Discrete Cosine Transform (DCT)
 - ▶ Wavelet Transform
 - ▶ Colour Quantisation
 - ▶ Chroma Subsampling
 - ▶ Fractal Compression

JPEG Standard

- ▶ JPEG is an image compression standard that was developed by the “Joint Photographic Experts Group” and accepted internationally in 1992
- ▶ JPEG uses a lossy image compression method called Discrete Cosine Transform (DCT)
 - ▶ One of frequency domain usages

JPEG Standard

- ▶ JPEG compression is based on 3 major observations
 1. Useful / Normal image pixels do not change their intensity values much



JPEG Standard

- ▶ JPEG compression is based on 3 major observations
 2. Human can notice the loss of low frequency components (i.e. background colours) better than the loss of very high frequency components (i.e. edges, noises)

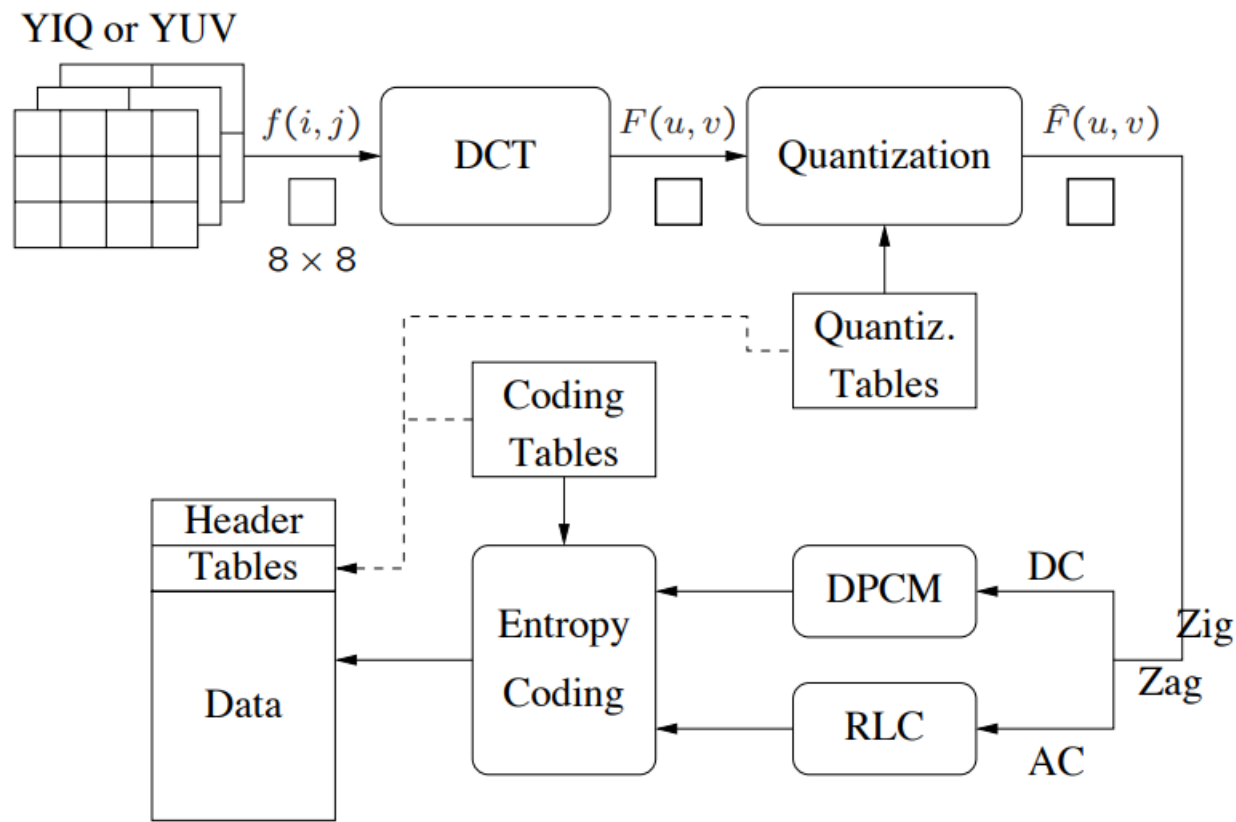
JPEG Standard

- ▶ JPEG compression is based on 3 major observations
 3. Human can tell the loss of grayscales better than colours



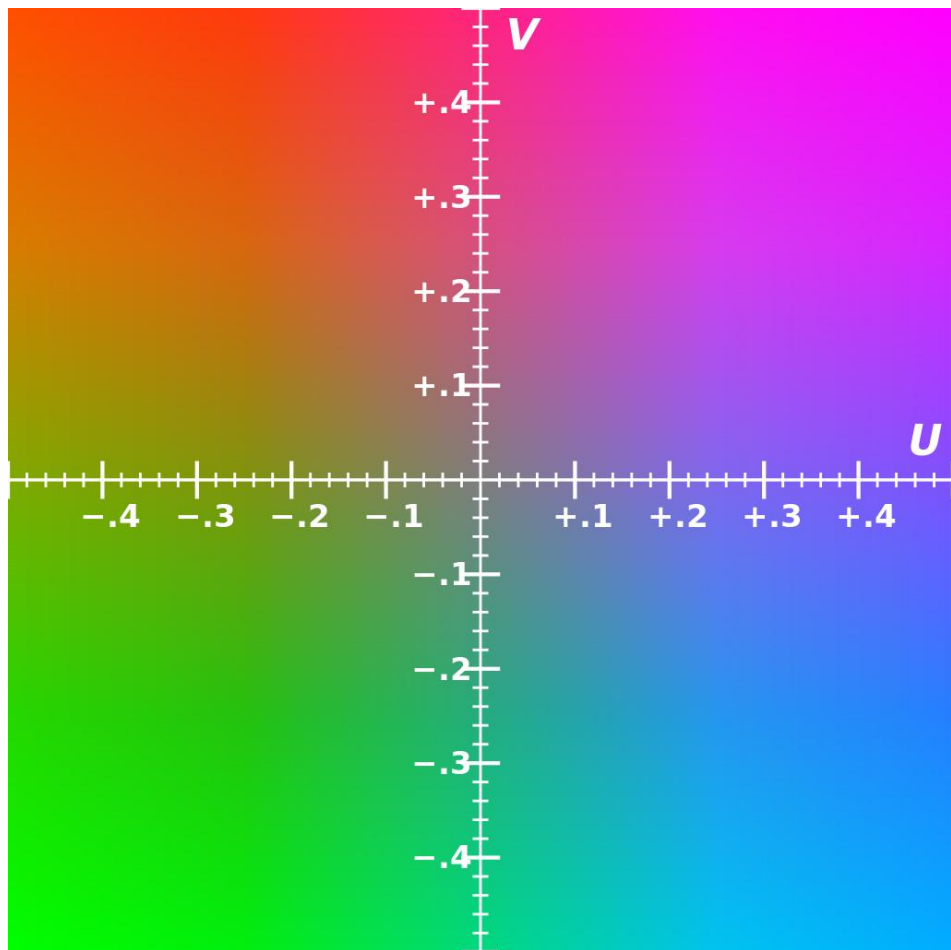
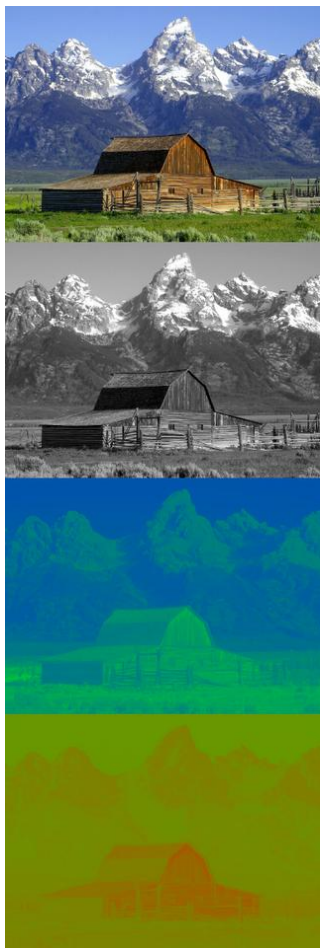
JPEG Image Compression

- ▶ Main steps of JPEG image compression are as follow
 - ▶ Transform RGB to YIQ or YUV and subsample colour.
 - ▶ DCT on image blocks.
 - ▶ Quantisation.
 - ▶ Zig-zag ordering and run-length encoding.
 - ▶ Entropy coding.



YUV

- ▶ YUV is one of colour models
- ▶ Y is luma (brightness), UV is colour gamut (Shades)
 - ▶ U for Blue – Yellow
 - ▶ V for Red – Cyan



Colour Subsampling



Y full resolution

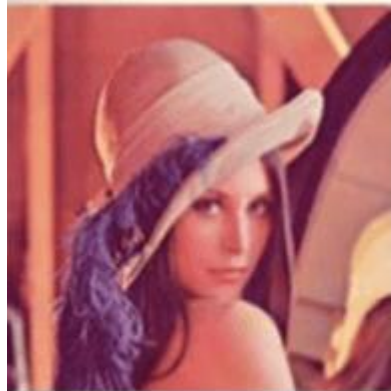


UV downsampled by
a factor of 20 in each
dimension



Discrete Cosine Transforms

- ▶ Each image is divided into 8×8 blocks, these are new pixels



Original image



Image split into 8x8 pixel blocks

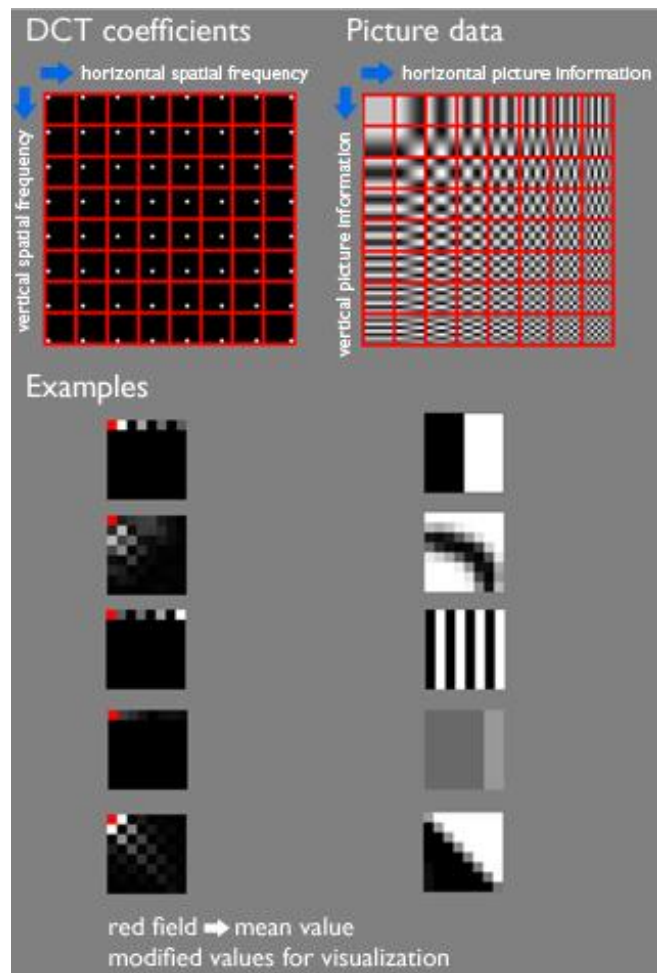
Discrete Cosine Transforms

- ▶ Recalling 2D DFT formula

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \left[\cos \left(2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right) \right) - j \sin \left(2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right) \right) \right]$$

- ▶ DCT is one form of DFT transforms which exclude the complex numbers (Sine part)

$$F(u, v) = \frac{a(u, v)}{4} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cos \left(\frac{(2x+1)u\pi}{16} \right) \cos \left(\frac{(2y+1)v\pi}{16} \right)$$



Quantisation

$$\hat{F}(u, v) = \text{round} \left(\frac{F(u, v)}{Q(u, v)} \right)$$

- ▶ $F(u, v)$ represents a DCT coefficient, $Q(u, v)$ is a “quantization matrix” entry, and $\hat{F}(u, v)$ represents the quantized DCT coefficients which JPEG will use in the succeeding entropy coding.



An 8×8 block from the Y image of 'Lena'

200	202	189	188	189	175	175	175
200	203	198	188	189	182	178	175
203	200	200	195	200	187	185	175
200	200	200	200	197	187	187	187
200	205	200	200	195	188	187	175
200	200	200	200	200	190	187	175
205	200	199	200	191	187	187	175
210	200	200	200	188	185	187	186

$f(i, j)$

515	65	-12	4	1	2	-8	5
-16	3	2	0	0	-11	-2	3
-12	6	11	-1	3	0	1	-2
-8	3	-4	2	-2	-3	-5	-2
0	-2	7	-5	4	0	-1	-4
0	-3	-1	0	4	1	-1	0
3	-2	-3	3	3	-1	-1	3
-2	5	-2	4	-2	2	-3	0

$F(u, v)$

32	6	-1	0	0	0	0	0
-1	0	0	0	0	0	0	0
-1	0	1	0	0	0	0	0
-1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

 $\hat{F}(u, v)$

199	196	191	186	182	178	177	176
201	199	196	192	188	183	180	178
203	203	202	200	195	189	183	180
202	203	204	203	198	191	183	179
200	201	202	201	196	189	182	177
200	200	199	197	192	186	181	177
204	202	199	195	190	186	183	181
207	204	200	194	190	187	185	184

 $\tilde{f}(i, j)$

512	66	-10	0	0	0	0	0
-12	0	0	0	0	0	0	0
-14	0	16	0	0	0	0	0
-14	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

 $\tilde{F}(u, v)$

1	6	-2	2	7	-3	-2	-1
-1	4	2	-4	1	-1	-2	-3
0	-3	-2	-5	5	-2	2	-5
-2	-3	-4	-3	-1	-4	4	8
0	4	-2	-1	-1	-1	5	-2
0	0	1	3	8	4	6	-2
1	-2	0	5	1	1	4	-6
3	-4	0	6	-2	-2	2	2

 $\epsilon(i, j) = f(i, j) - \tilde{f}(i, j)$



Another 8×8 block from the Y image of 'Lena'

```

70 70 100 70 87 87 150 187
85 100 96 79 87 154 87 113
100 85 116 79 70 87 86 196
136 69 87 200 79 71 117 96
161 70 87 200 103 71 96 113
161 123 147 133 113 113 85 161
146 147 175 100 103 103 163 187
156 146 189 70 113 161 163 197

```

$f(i, j)$

```

-80 -40 89 -73 44 32 53 -3
-135 -59 -26 6 14 -3 -13 -28
47 -76 66 -3 -108 -78 33 59
-2 10 -18 0 33 11 -21 1
-1 -9 -22 8 32 65 -36 -1
5 -20 28 -46 3 24 -30 24
6 -20 37 -28 12 -35 33 17
-5 -23 33 -30 17 -5 -4 20

```

$F(u, v)$

-5	-4	9	-5	2	1	1	0
-11	-5	-2	0	1	0	0	-1
3	-6	4	0	-3	-1	0	1
0	1	-1	0	1	0	0	0
0	0	-1	0	0	1	0	0
0	-1	1	-1	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

 $\hat{F}(u, v)$

70	60	106	94	62	103	146	176
85	101	85	75	102	127	93	144
98	99	92	102	74	98	89	167
132	53	111	180	55	70	106	145
173	57	114	207	111	89	84	90
164	123	131	135	133	92	85	162
141	159	169	73	106	101	149	224
150	141	195	79	107	147	210	153

 $\tilde{f}(i, j)$

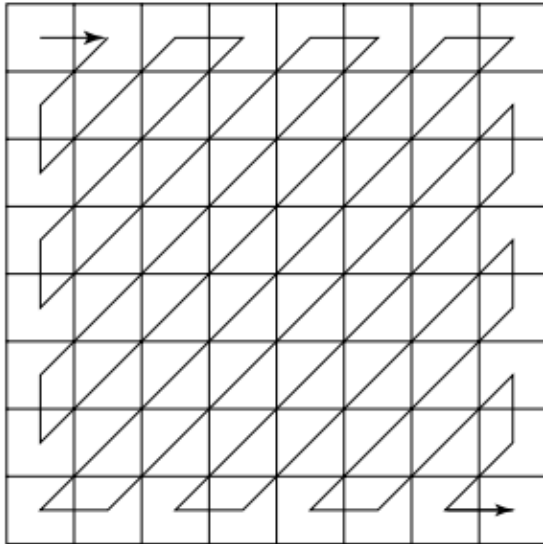
-80	-44	90	-80	48	40	51	0
-132	-60	-28	0	26	0	0	-55
42	-78	64	0	-120	-57	0	56
0	17	-22	0	51	0	0	0
0	0	-37	0	0	109	0	0
0	-35	55	-64	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

 $\tilde{F}(u, v)$

0	10	-6	-24	25	-16	4	11
0	-1	11	4	-15	27	-6	-31
2	-14	24	-23	-4	-11	-3	29
4	16	-24	20	24	1	11	-49
-12	13	-27	-7	-8	-18	12	23
-3	0	16	-2	-20	21	0	-1
5	-12	6	27	-3	2	14	-37
6	5	-6	-9	6	14	-47	44

 $\epsilon(i, j) = f(i, j) - \tilde{f}(i, j)$

Run-length Coding (RLC) on AC coefficients



512	66	-10	0	0	0	0	0
-12	0	0	0	0	0	0	0
-14	0	16	0	0	0	0	0
-14	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

$\tilde{F}(u, v)$



IMAGE RECONSTRUCTION

Image Reconstruction

- ▶ Image Reconstruction is the process of creating a new image using information from several sources
- ▶ The more information there is, the better result is
 - ▶ Image Reconstruction from Projections
 - ▶ Image Reconstruction from Exposures



Image Reconstruction from Projections

Image Reconstruction from Projections

- ▶ An image can be reconstructed if there are enough series of projections



- ▶ One of the most-widely used application is Computed Tomography (CT scan)

Image Reconstruction from Projections

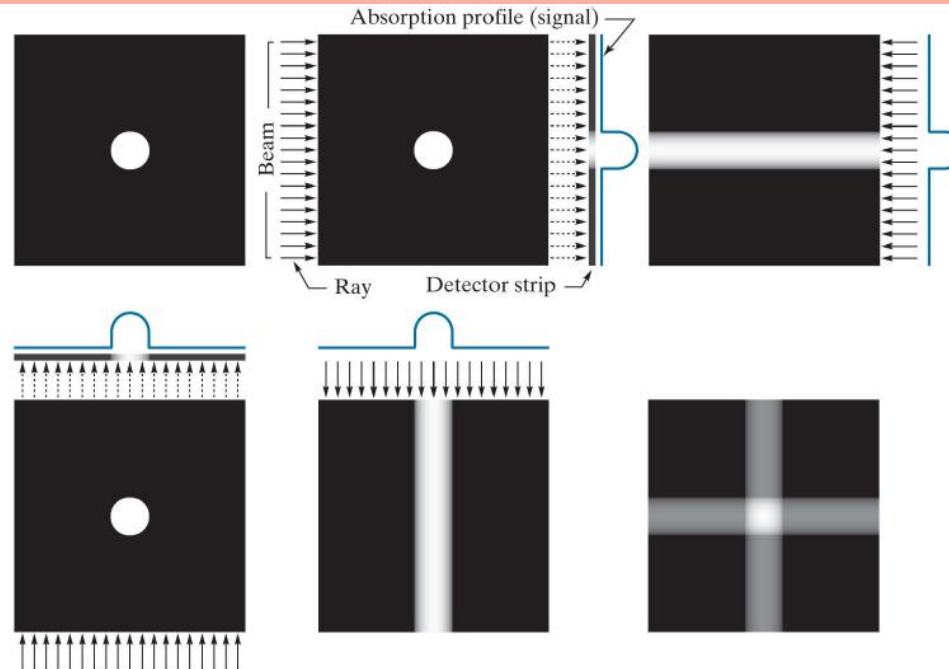


Image Reconstruction from Projections

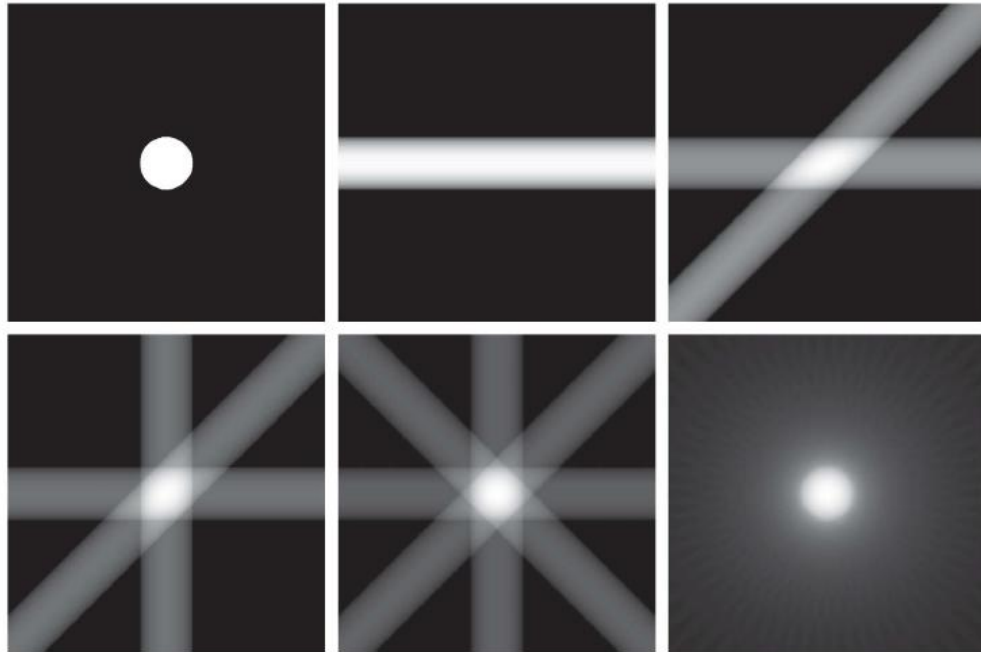
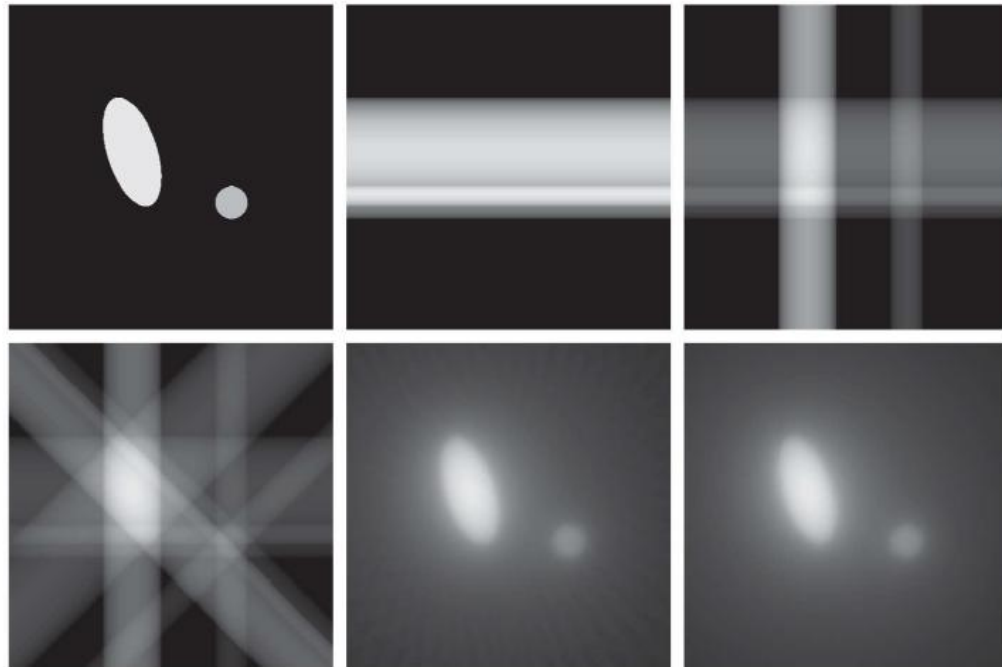
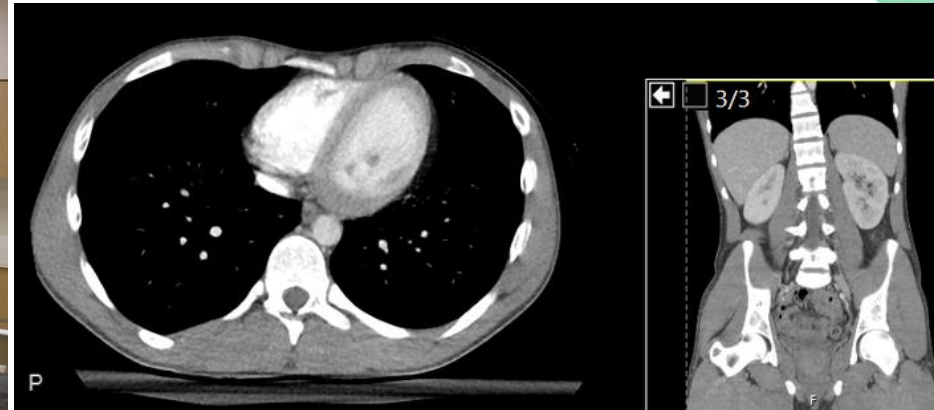


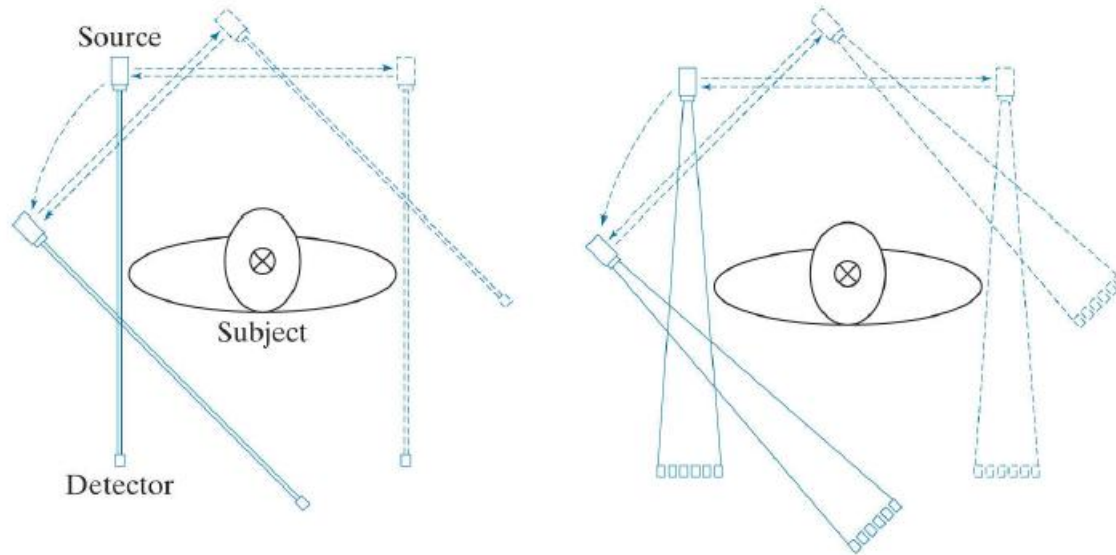
Image Reconstruction from Projections



CT Scanner



CT Scanner



CT Scanner

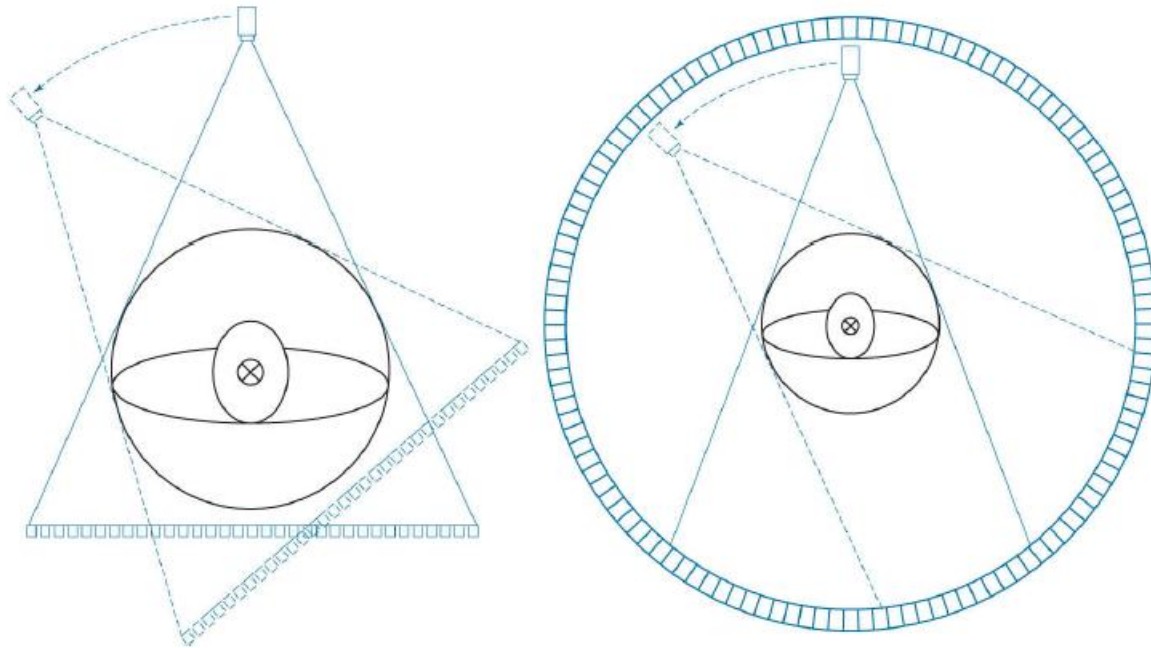
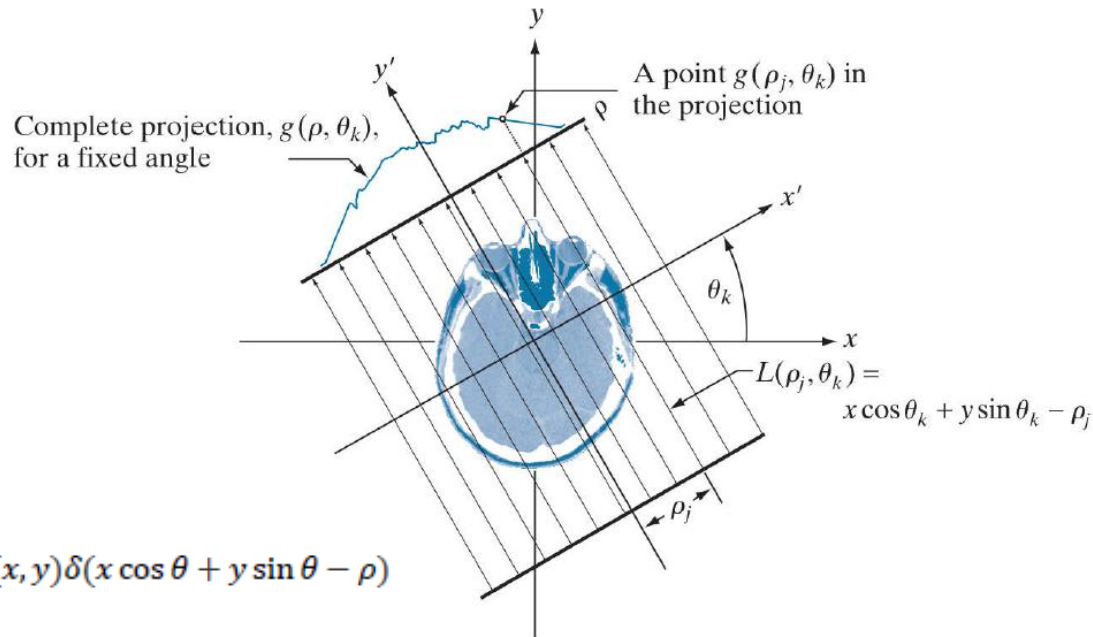


Image Reconstruction from Projections



$$g(\rho, \theta) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \delta(x \cos \theta + y \sin \theta - \rho)$$

Image Reconstruction from Projections

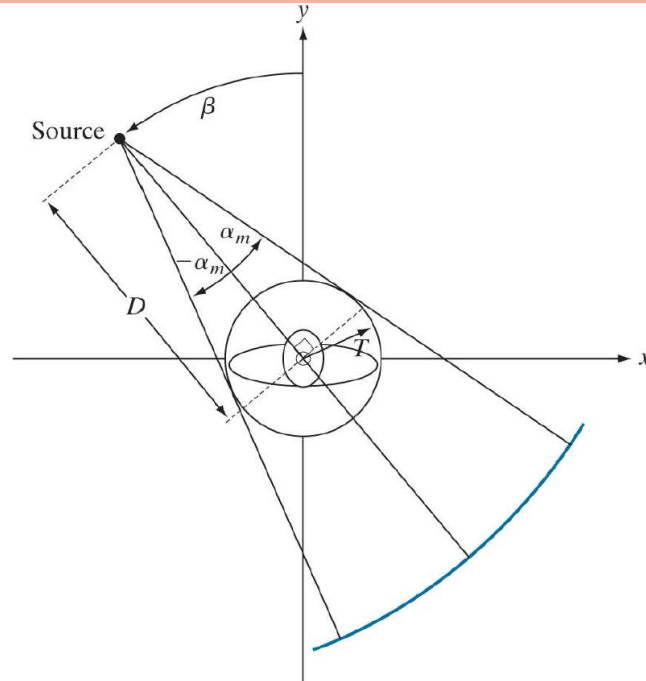


Image Reconstruction from Projections

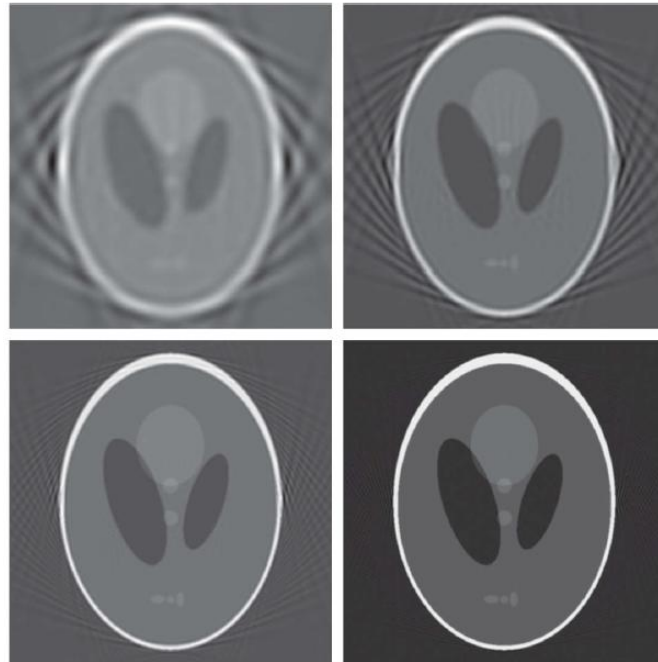




Image Reconstruction from Exposures

High-dynamic-range imaging

- ▶ Normally, brightness of digital images is clamped within 0 – 255 because of the limitation of computer screen or printing (Low-dynamic-range). But, in fact, the brightness can fall above or below the range. This is HDR image
- ▶ The HDR images can be acquired by two sources
 - ▶ Special image sensors
 - ▶ Merging multiple normal images into one HDR image
- ▶ High-dynamic-range imaging (HDR, HDRI) is the set of digital image processing techniques to recreate an image using set of images with greater range of luminosity

High-dynamic-range imaging



High-dynamic-range imaging

- ▶ The processes of merging several images to create an HDR image is called **Tone Mapping**
- ▶ There are several types of tone mapping (called tone-mapping operator – TMO) which are different in the processes. They are generally divided into two types
 - ▶ Global tone-mapping analysis
 - ▶ Fast but use the general exposures of image. So, the final result will be flat
 - ▶ Local tone-mapping analysis
 - ▶ Slower but separate bright areas from the shadow area. The final result will be better

High-dynamic-range imaging



High-dynamic-range imaging



High-dynamic-range imaging

- ▶ For more information – visit
- ▶ <http://pages.cs.wisc.edu/~lizhang/courses/cs766-2008f/projects/hdr/Ashikhmin2002ATM.pdf>



End of Week 7